UpToDate®
Official reprint from UpToDate®
www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Clinical features, diagnosis, and management of relapsing fever

AUTHOR: Alan G Barbour, MD
SECTION EDITORS: Daniel J Sexton, MD, Sheldon L Kaplan, MD
DEPUTY EDITOR: Jennifer Mitty, MD, MPH

All topics are updated as new evidence becomes available and our peer review process is complete.

Literature review current through: **Jul 2026.**
This topic last updated: **May 29, 2024.**

INTRODUCTION

Relapsing fever, caused by spirochetes of the *Borrelia* genus, is an arthropod-borne infection that occurs in two major forms: tick-borne (TBRF) and louse-borne (LBRF) [1,2]:

- TBRF is a zoonosis that is enzootic in many countries. The two main *Borrelia* species involved in North America are *Borrelia hermsii* (in the mountainous west) and *Borrelia turicatae* (in the southwest and south-central region) [3]. Other tick-borne species cause relapsing fever on other continents.
- LBRF is caused by *Borrelia recurrentis*. It is principally a disease seen in the developing world or among refugees from developing countries [1,4]. It is spread from person to person by the body louse and can result in large epidemics.

As the name implies, relapsing fever is characterized by recurrent episodes of fever, which accompany spirochetemia. The disease relapses are due to antigenic variation by the spirochetes.

The clinical manifestations, diagnosis, treatment, and prevention of relapsing fever will be reviewed here. Discussions of the microbiology, pathogenesis, and epidemiology of relapsing fever, as well as *Borrelia miyamotoi*, a *Borrelia* species related to the agents of relapsing fever, are found elsewhere. (See "Microbiology, pathogenesis, and epidemiology of relapsing fever" and "Borrelia miyamotoi infection".)

CLINICAL MANIFESTATIONS

Course of fever — Relapsing fever presents with the sudden onset of fever, punctuated by an intervening afebrile period, and then recurrent fevers. The temperature is usually above 39°C, and may be as high as 43°C. The incubation period between the exposure and the first fever is between 3 and 12 days.

The first fever episode ends by a crisis phase, which lasts for approximately 15 to 30 minutes and consists of rigors, a further elevation in temperature, and an increase in pulse and blood pressure. The crisis phase is followed by profuse diaphoresis, falling temperature, and hypotension, which usually persist for several hours. Mortality from untreated relapsing fever is most common during the crisis and its immediate aftermath. (See 'Prognosis' below.)

In both tick-borne and louse-borne relapsing fever, the interval between fevers ranges from 4 to 14 days. However, the pattern of fever varies depending upon the causative agent:

- In tick-borne relapsing fever (TBRF), multiple febrile periods each last from one to three days (figure 1) [3,5,6].
- In louse-borne relapsing fever (LBRF), the first episode of fever is unremitting for three to six days; it is typically followed by a single milder episode [1].

Following the initial symptomatic infection, and during subsequent relapses, a succession of *Borrelia* serotypes appear in the blood. Each serotype is individually defined by antigenically distinct surface proteins [7]. (See "Microbiology, pathogenesis, and epidemiology of relapsing fever", section on 'Variable major proteins'.)

Other signs and symptoms — In addition to recurring fevers, patients with relapsing fever may present with a variety of other signs and symptoms. Some may be seen in patients with both TBRF and LBRF, whereas others are typically associated with a specific pathogen:

- **Generalized** – Patients with relapsing fever generally present with nonspecific symptoms (eg, headache, myalgia, arthralgia, shaking chills, and abdominal complaints). These symptoms may accompany the first and subsequent febrile episodes, although the first episode of illness tends to be the most severe and may initially be diagnosed as sepsis [8]. In one report, nausea with vomiting was the predominant symptom in febrile children 11 years old or younger [9].

- **Neurologic** – Patients with relapsing fever may develop neurologic complications. Neurologic manifestations, such as delirium, apathy, stupor, dizziness, or, rarely, coma, can be seen in both TBRF and LBRF. In LBRF, these neurologic manifestations are usually secondary to spirochetemia and systemic inflammation, rather than direct invasion of organisms into the central nervous system (CNS) [10]; however, meningeal involvement with abnormal cerebrospinal fluid may occur [11].

  Localized neurologic signs (eg, hemiplegia, facial palsy, myelitis, meningitis, and radiculopathy) are more common with TBRF and are due to direct invasion of spirochetes [10,12]. Visual loss may be permanent in patients who develop bilateral iridocyclitis or panophthalmitis. Localized neurologic manifestations are more likely to present during subsequent, rather than the initial febrile period. As an example, cranial neuritis typically presents in the second or third febrile episode, not the first.

- **Cardiopulmonary** – Patients may also experience cardiac or respiratory complications. Myocarditis appears to be common in both TBRF and LBRF, and has been reported as a prominent feature in fatal cases [13]. Pulmonary symptoms range from acute respiratory distress syndrome, which is typically seen in patients with TBRF [14], to a nonproductive cough in patients with LBRF.

- **Hematologic** – Hematologic complications can be seen in patients with LBRF. Such patients can present with a bleeding disorder, which is probably the consequence of thrombocytopenia, impaired hepatic production of clotting factors, and/or blockage of small vessels by aggregates of spirochetes, erythrocytes, and platelets.

Physical examination — Exam findings seen in patients with both TBRF and LBRF include splenomegaly that may be accompanied by abdominal or left shoulder pain, and hepatomegaly, which is seen in the majority of patients with LBRF and in about 10 percent of patients with TBRF. In patients with myocarditis, a gallop on cardiac auscultation is commonly detected. Although patients with relapsing fever frequently complain of polyarthralgia, enlarged, painful joints are unusual.

Other exam findings, such as localized neurologic findings, are more common in TBRF than LBRF. Bell's palsy (unilateral or bilateral) and deafness from seventh or eighth cranial nerve involvement are the most common forms of cranial neuritis in TBRF. See above.

By contrast, patients with LBRF may have findings suggestive of a bleeding disorder, such as epistaxis, petechiae, and ecchymoses [15-17]. Subarachnoid hemorrhage has been observed, perhaps due to disseminated intravascular coagulation. See above.

General laboratory findings — Nonspecific laboratory findings are generally seen in patients with relapsing fever. As examples:

- A mild-to-moderate normocytic anemia is common, but frank hemolysis and hemoglobinuria do not occur.

- Leukocyte counts are usually in the normal range or only slightly elevated; however, there can be leukopenia during the crisis.
- Platelet counts can fall below 50,000/microL.
- The erythrocyte sedimentation rate may be elevated, sometimes above 100 mm/hour [9].
- Hepatitis can occur with elevated serum concentrations of unconjugated bilirubin and aminotransferases.
- The prothrombin and partial thromboplastin times may be moderately prolonged.
- C-reactive protein and procalcitonin levels may be elevated, sometimes markedly [8,11,18].

When analysis of the cerebrospinal fluid (CSF) is performed (eg, when there are signs of meningitis or meningoencephalitis), the CSF examination usually reveals a mononuclear pleocytosis and/or mildly to moderately elevated protein levels [10]. Glucose concentrations are usually not depressed.

Other studies — In patients with myocarditis, the electrocardiogram may reveal a prolonged QTc interval. Some patients may also have cardiomegaly and pulmonary edema on chest radiograph.

PREGNANCY

Pregnant women with relapsing fever tend to have a more severe and prolonged illness [19]. In addition, transplacental transmission of the infection can occur. Relapsing fever during pregnancy frequently leads to premature labor, abortion, or stillbirth; however, congenital malformations as a consequence of relapsing fever have not been reported. A newborn may also be infected shortly after birth, and neonates with relapsing fever are at increased risk for complications [20]. In a meta-analysis of 206 pregnancies reported in 14 studies, the incidence of an adverse pregnancy outcome for LBRF was 71 percent [21].

A murine model of gestational relapsing fever infection found that infection during pregnancy causes intrauterine growth retardation, placental damage and inflammation, impaired fetal circulation, and decreased maternal hemoglobin levels [22]. In a murine model, spirochetes frequently crossed the maternal-fetal barrier, leading to infection of the fetus. The model closely parallels human infection and provides potential mechanisms to account for the observed human pathology.

DIAGNOSIS

Diagnostic approach — Relapsing fever should be considered in patients who have both of the following:

- Characteristic relapsing fevers, especially if the recurrent fevers are accompanied by the crisis phenomenon. (See 'Course of fever' above.)
- An epidemiologic history of exposure to soft-bodied ticks or body lice in geographic areas where louse-borne relapsing fever (LBRF) or tick-borne relapsing fever (TBRF) is endemic (figure 2). LBRF may also occur in household or refugee camp contacts of a recently-arrived traveler or migrant [4]. (See "Microbiology, pathogenesis, and epidemiology of relapsing fever", section on 'Epidemiology'.)

However, patients commonly seek care during the initial bacteremia and present only with an undifferentiated febrile illness. In areas where TBRF or LBRF is endemic, local practitioners may be alert for relapsing fever, but if the illness begins in someone who is distant from the exposure site, it may not be suspected.

To evaluate the patient for relapsing fever, we first perform thin and thick smears of blood to see if the organism can be identified [23]. If the organism is not identified on smear, but there is a high clinical suspicion for disease,

polymerase chain reaction (PCR) testing should be performed. (See 'Direct visualization procedures' below and 'Polymerase chain reaction' below.)

Patients with suspected central nervous system (CNS) involvement should have a lumbar puncture. Cerebrospinal fluid (CSF) should be sent for microscopic examination and PCR testing. In patients with meningitis or meningoencephalitis, microscopic examination of the CSF may reveal spirochetes, but their presence in the subarachnoid space is more likely revealed by PCR. (See 'Direct visualization procedures' below and 'Polymerase chain reaction' below.)

Culture and serologic techniques are less helpful when trying to diagnose relapsing fever. Culture can be difficult to perform and is not routinely available. The primary indication for serologic testing is retrospective confirmation of a diagnosis. Serologic testing for relapsing fever is not indicated for patients with long-standing symptoms or disability. (See 'Cultures' below and 'Serologies' below.)

Diagnostic methods

Direct visualization procedures — Direct visualization includes identifying the organism in blood or in tissue specimens.

- **Identifying the organism in blood** – Thin and thick smears of blood should be obtained to see if the organism can be visualized. The optimum time to obtain blood is between the fever's onset and its peak. Spirochetes are usually unable to be visualized in blood once the temperature is declining or back to the normal range in the absence of antipyretics. With current automated procedures for blood smear analysis, a request for a manual examination of the blood smear must be made.

  - On thin smear, Giemsa or Wright stains typically reveal spirochetes if the concentration of microorganisms is greater than 10^5/mL (picture 1). As many as 200 oil immersion fields should be viewed before judging the smear to be negative.

  - A thick smear is able to detect at least one log fewer organisms (ie, 10^4/mL of blood). To perform a thick smear, the slide should first be treated with 0.5 percent acetic acid to lyse hemoglobin, and then stained with Giemsa or Wright stain. The acetic acid treatment can be omitted if the smear is stained with acridine orange [13], which binds to nucleic acids in the spirochetes. The slide should then be examined microscopically under ultraviolet light.

  - For both thin and thick smears, direct or indirect immunofluorescence is another procedure that can be used to visualize spirochetes [24]. A fluorescein-labeled polyclonal antibody to *Borrelia* species is commercially available and is useful for detecting relapsing fever spirochetes in blood or in tissues [25].

  Other procedures have also been used to visualize the organism. As an example, examination of the buffy coat and overlying plasma can detect as few as 10^3 spirochetes/mL of blood. Further enrichment for spirochetes in the blood may be achieved with a two-step centrifugation procedure [26].

  Another diagnostic maneuver is the use of a wet mount technique. Uncentrifuged plasma or the concentrated buffy coat is examined by phase contrast or dark field microscopy, and coiling, uncoiling, and bending movements of the spirochetes can be visualized as they swim among the erythrocytes.

  A smear or wet mount properly evaluated can be diagnostic since *Borrelia* spp are an unlikely contaminant. However, in inexperienced hands, stain artifacts in blood smears may produce a false diagnosis.

- **Identifying the organism in tissue specimens** – The spirochetes can be detected in tissue sections with silver stains, such as Warthin-Starry or modified Dieterle, or by immunofluorescence with conjugated antibodies [27,28].

Polymerase chain reaction — PCR testing is useful for the diagnosis of relapsing fever [29-31]. It is available at some public health departments and commercial reference laboratories. Samples can also be sent to the United States Centers for Disease Control and Prevention (CDC) through State Health Department laboratories.

The PCR assay is more sensitive than a blood smear, especially when numbers of spirochetes in the blood are low between fever episodes.PCR can be performed on samples of anticoagulated blood, blood clot, platelet-rich plasma, or CSF. Tissue from the liver, brain, or spleen obtained post-mortem may contain the spirochetes when they are no longer in blood.

The most readily available PCR assays are at commercial reference laboratories; these assays identify the genus as *Borrelia* but not the species of *Borrelia*. Consequently, they cannot distinguish the agents of either TBRF or LBRF from *B. miyamotoi* (see "Borrelia miyamotoi infection"). With genus-specific PCR the species of *Borrelia* can be identified by follow-up DNA sequencing of the PCR product [32], but this is not routinely done. Direct identification of species in a patient specimen has been carried out by multiplex PCR assay [33] or metagenomic DNA sequencing [34], both of which have limited availability.

Cultures — The diagnosis of relapsing fever can be established by culturing the organism [35-38]. However, only a few laboratories in the world perform cultures for the organisms that cause relapsing fever, and some species, including *B. recurrentis*, are difficult to cultivate. An alternative is isolation of the spirochete from a patient specimen by inoculation of a laboratory mouse with citrated blood [39].

Serologies — Serologic testing is not typically used to diagnose relapsing fever. Serology is most suitable to confirm a diagnosis after a patient has been treated [40], or when doing serosurveys of populations to assess for evidence of past infection [41]. There is no evidence that relapsing fever can develop into a chronic or persistent infection, even in the absence of specific therapy. Thus, serologic testing for relapsing fever is not indicated for patients with long-standing symptoms or disability.

Serologic assays in the format of indirect immunofluorescence of whole cells of a relapsing fever *Borrelia* species are available at some commercial reference laboratories. However, the cells for these assays are cultivated in the laboratory and may not include some of the immunodominant antigens to which the patient is responding. In addition, there may be cross-reactivity of several of the proteins of the relapsing fever *Borrelia* cells with those of the *Borreliella* species that cause Lyme disease [25], so a positive result may not be attributable to relapsing fever. (See "Microbiology, pathogenesis, and epidemiology of relapsing fever", section on 'Pathogenesis'.)

An alternative assay can detect antibodies to the GlpQ protein, which is produced by relapsing fever *Borrelia* species, including *B. miyamotoi*, but not by the agents of Lyme disease [42,43]. Commercially available assays for antibodies to the GlpQ protein of *B. miyamotoi* yield positive results with sera from patients with other forms of relapsing fever on the basis of the extensive cross-reactivity between GlpQ proteins of different species [44-46]. (See "Borrelia miyamotoi infection".)

DIFFERENTIAL DIAGNOSIS

In an emergency department, the provisional diagnosis for unsuspected relapsing fever is commonly sepsis of unknown origin. Suspicion for relapsing fever depends on the patient's travel history, as well as residential, occupational, and recreational exposures. As an example, household pets are not a risk factor for tick-borne relapsing fever (TBRF) in North America, but domestic animals, such as pigs, housed next to or inside the family residence may be hosts for soft ticks in less-developed areas. (See "Microbiology, pathogenesis, and epidemiology of relapsing fever".)

Conditions that mimic some manifestations of relapsing fever or are transmitted by ticks include:

- Malaria, in particular falciparum malaria in areas where both diseases are endemic (see "Malaria: Clinical manifestations and diagnosis in nonpregnant adults and children")

- Leptospirosis (see "Leptospirosis: Epidemiology, microbiology, clinical manifestations, and diagnosis")
- Typhoid fever (see "Enteric (typhoid and paratyphoid) fever: Epidemiology, clinical manifestations, and diagnosis", section on 'Introduction')
- Tularemia (see "Tularemia: Clinical manifestations, diagnosis, treatment, and prevention")
- Colorado tick fever (see "Arthropod-borne encephalitides", section on 'Colorado tick fever virus')
- Rickettsioses (see "Other spotted fever group rickettsial infections" and "Epidemiology, clinical manifestations, and diagnosis of Rocky Mountain spotted fever")
- Dengue (see "Dengue virus infection: Clinical manifestations and diagnosis")
- Rat bite fever (see "Rat bite fever")

While the infections listed above share some clinical manifestations with relapsing fever, many can usually be excluded on epidemiologic grounds. As an example, while patients with the mosquito-borne viral infection dengue may present with a "saddleback" fever pattern that suggests relapsing fever, there are few areas in the world where the distributions of dengue and TBRF overlap. However, there are areas of overlap, where TBRF and malaria are seen (eg, rural areas of sub-Saharan Africa) and where TBRF due to *B. hermsii* and Colorado tick fever can occur (eg, the mountains of western North America).

In patients with suspected louse-borne relapsing fever (LBRF), it is also important to consider illnesses that may accompany the louse-borne form. This includes infections that occur under the same adverse living conditions that foster the endemicity of body louse infestations (eg, measles, typhoid, and tuberculosis) as well as other infections that are transmitted by lice, such as epidemic typhus and trench fever (*Bartonella quintana* bacteremia). (See "Epidemic typhus" and "*Bartonella quintana* infections: Clinical features, diagnosis, and treatment".)

By contrast, the *Ornithodoros* soft ticks, which are vectors of TBRF, are not known to transmit other infections to humans.

TREATMENT

Indications — Antimicrobial therapy should be administered to all patients diagnosed with relapsing fever. With prompt antibiotic treatment, the mortality rate is substantially reduced. (See 'Prognosis' below.)

Empiric therapy is also reasonable for patients with a suspected diagnosis based upon both characteristic clinical findings and a history of recent exposure to either soft ticks (eg, an overnight stay in a mountain cabin, exploration of a cave) or body lice (eg, housing in a refugee camp). (See 'Diagnostic approach' above.)

It can be difficult to make a definitive diagnosis of relapsing fever since a negative smear of a blood sample obtained after the initial defervescence does not exclude the diagnosis, and the results of polymerase chain reaction (PCR) testing may not be available for days. (See 'Diagnostic methods' above.)

Antimicrobial regimens — The approach to treatment depends upon the causative agent. (See 'Tick-borne relapsing fever' below and 'Louse-borne relapsing fever' below.)

The minimum inhibitory concentrations (MICs) of penicillin G, tetracyclines, and macrolides for *Borrelia* spp are generally <1 mcg/mL [38,47-49]. Relapsing fever *Borrelia* spp are susceptible in vitro to second- and third-generation (but not first-generation) cephalosporins as well as chloramphenicol, but there is less clinical experience with these antibiotics. *Borrelia* species are relatively resistant to rifampin, sulfonamides, fluoroquinolones, metronidazole, and aminoglycosides.

A longer duration of treatment is needed for tick-borne relapsing fever (TBRF) compared with louse-borne relapsing fever (LBRF), since the relapse rate of TBRF is 20 percent or higher after single-dose treatments [5,50,51]. This may be due to invasion of TBRF spirochetes in the brain, which serves as a sanctuary from where the spirochetes can reinvade the blood once antibiotic levels have fallen [5,49].

Tick-borne relapsing fever — There are limited published data to support a specific treatment strategy for TBRF. Although TBRF is generally more common than LBRF, the sporadic nature of TBRF and its tendency to occur in rural or remote regions mean that there are few opportunities for clinical trials or extended case series. Nevertheless, cumulative clinical experience provides useful guidance.

The majority of patients with relapsing fever present with severe symptoms and require hospitalization. In general, we prefer to initiate therapy with an intravenous (IV) beta-lactam, such as penicillin or ceftriaxone (table 1). Decades of experience support the use of penicillin as treatment of relapsing fever. However in vitro data suggest that amoxicillin (and, by implication, ampicillin) may be less effective than penicillin or ceftriaxone [52], and therefore, we do not use ampicillin.

Additional considerations regarding treatment depend upon the presence of central nervous system (CNS) disease and whether the patient is pregnant (table 1).

- **Patients without CNS disease** – The duration of therapy for TBRF is 10 days for nonpregnant patients without CNS disease, and patients can typically transition to oral therapy after they are clinically stable. We prefer to switch the patient to an oral tetracycline (eg, doxycycline), if possible. Oral tetracyclines have good CNS penetration, and thus may reduce the risk of a relapse occurring via spirochetes re-entering the blood from the CNS once antibiotic levels have fallen.

  For patients without CNS disease who cannot take a beta-lactam as initial therapy, a tetracycline (orally or intravenously), such as doxycycline, should be used. Macrolides (eg, azithromycin) are typically administered if the patient cannot tolerate a beta-lactam or a tetracycline.

  Special considerations regarding the use of tetracyclines in children are discussed below. (See 'Special considerations for pregnant women and young children' below.)

- **Patients with CNS involvement** – For patients with CNS involvement, the duration is extended to up to 14 days. IV therapy with penicillin or ceftriaxone should be continued for the entire duration. The one exception is patients who present with isolated Bell's palsy. Such patients can transition to oral therapy with a tetracycline after they are clinically stable.

  For patients with CNS disease, all efforts should be made to administer penicillin or ceftriaxone. However, if that is not possible, a tetracycline (eg, doxycycline) is the preferred alternative agent since tetracyclines have good CNS penetration. The use of tetracyclines in children is discussed below. (See 'Special considerations for pregnant women and young children' below.)

- **Pregnant women** – IV therapy with penicillin or a beta-lactam should be continued for 14 days in pregnant women (regardless of CNS disease), given the greater risk of mortality for the mother and fetus or neonate. (See 'Pregnancy' above.)

  Similar to patients with CNS disease, all efforts should be made to administer penicillin or ceftriaxone. However, if this is not possible, the choice of an alternative agent depends upon the risks of therapy. (See 'Special considerations for pregnant women and young children' below.)

In resource-limited settings, several days of IV therapy may not be feasible. For such patients, a tetracycline is preferred. If a tetracycline is not available or tolerated, the next best option is a macrolide. If first-line options are not possible, oral penicillin can be administered in divided doses (eg, penicillin V potassium [VK] 500 mg or 12.5 mg/kg every eight hours) as long as CNS involvement is not suspected.

After therapy is initiated, all patients (even those who receive oral therapy) should be monitored for a Jarisch-Herxheimer reaction. (See 'Jarisch-Herxheimer reactions' below.)

Louse-borne relapsing fever — Most patients with LBRF can be treated with a single dose of intramuscular penicillin or a single dose of a tetracycline. Patients should be observed for a Jarisch-Herxheimer reaction after treatment. (See 'Jarisch-Herxheimer reactions' below.)

- A single dose of intramuscular penicillin G procaine (400,000 to 800,000 units) can be used for adults [53-56]. For children, we administer intramuscular penicillin G procaine (200,000 to 400,000 units).
- Tetracycline can be administered as a single dose of oral or IV doxycycline (for adults the dose of doxycycline is 200 mg; for children the dose is 5 mg/kg up to a maximum dose of 200 mg). If doxycycline is not available, a single dose of tetracycline (500 mg) can be administered to nonpregnant adults. Considerations for pregnant women and young children are discussed below. (See 'Special considerations for pregnant women and young children' below.)
- For those unable to take a beta-lactam or a tetracycline, a single dose of a macrolide, such as erythromycin base (500 mg orally for adults; 12.5 mg/kg [maximum dose 500 mg] orally for children) can be used. It is likely that azithromycin (eg, 500 mg for adults; 10 mg/kg [maximum dose 500 mg] for children), which is better tolerated, is equally effective.

A meta-analysis of antibiotic regimens for LBRF in adults found that there was no difference in mortality rates between those receiving tetracycline (orally or intravenously) or penicillin G-based therapy, although tetracycline was marginally superior regarding the time interval from treatment to clearance of fever (mean difference 3.5 hours, 95% CI 3.06–3.97) and relapse rate [57]. The recurrence rate in patients receiving any of these agents is <5 percent.

Special considerations for pregnant women and young children — The approach to treatment in persons who are pregnant and in young children is described in the table (table 1). Similar to others, the preferred regimens typically include penicillin or a third-generation cephalosporin. These agents are not associated with known teratogenic effects.

If there is a contraindication to beta-lactams, doxycycline is a reasonable alternative. Doxycycline has not been associated with serious adverse events in pregnant women or children <8 years old. However, the risk appears greater if other tetracyclines are administered. Thus, a macrolide may be preferred if neither a beta-lactam nor doxycycline can be used. These decisions should be made on a case-by-case basis.

Tetracyclines other than doxycycline are generally contraindicated in pregnancy because of the risk of hepatotoxicity in the mother and adverse effects on fetal bone and teeth (eg, permanent discoloration of deciduous teeth from in utero exposure in the second and third trimesters [58], incorporation into fetal long tubular bones with transient inhibition of growth [59]). In addition, tetracyclines other than doxycycline caused dental staining when administered to children younger than eight years. However, these events are extremely rare with doxycycline, and observational studies support the relative safety of doxycycline compared with older tetracyclines in both pregnancy and in children [60,61].

Jarisch-Herxheimer reactions — Patients with relapsing fever often experience a moderate-to-severe Jarisch-Herxheimer reaction (JHR), typically starting within one hour and peaking at four hours after initiating antibiotics [62,63]. This reaction has been reported in approximately 60 percent of patients being treated for LBRF [21,64] and in up to 54 percent of patients being treated for TBRF. Both penicillin and tetracycline can induce a JHR [57] and, most likely, macrolides can as well.

Patients with either LBRF or TBRF should be observed for four to six hours after the first antibiotic dose. Signs and symptoms of the JHR include rigors, myalgias, further rise in the body temperature, increased respiratory rate, and

hypotension lasting up to 24 hours. There may cardiac dysfunction with elevation of troponin levels in the blood [63]. The severity of the JHR is a function of the numbers of spirochetes in the blood [65].

If treatment of an otherwise undifferentiated febrile illness with a beta-lactam, tetracycline, or macrolide antibiotic results in an unexpected worsening of the patient's condition, relapsing fever should be considered as a possible diagnosis if epidemiologic risk factors are present. However, a JHR is not specific for relapsing fever. As an example, a moderate-to-severe JHR may occur after initial antibiotic doses for treatment of secondary syphilis or leptospirosis. A JHR may be misdiagnosed as an anaphylactic reaction to the antibiotic.

The occurrence of a JHR should not lead to a change in antibiotics, since the presence of a JHR is indicative of a therapeutic response. Spirochetes are not detectable in blood smears of most patients within a few hours of the first dose of an effective antibiotic. Patients who develop a JHR generally improve after the first day of treatment, but antipyretics may reduce the severity of symptoms and the duration of the reaction. Patients who develop a severe JHR may need more intensive management (frequently IV fluid resuscitation and, less commonly, vasopressors).

Proinflammatory cytokines, especially tumor necrosis factor (TNF)-alpha and interleukin (IL)-6 and IL-8, have been implicated in the pathogenesis of JHR [64]. In one study of 49 patients with proven LBRF, pretreatment with anti-TNF-alpha antibodies reduced the incidence of rigors from 90 percent in placebo recipients to 50 percent, and the mean increase in temperature from 1.5°C to 0.8°C [56]. However, administration of pentoxifylline or recombinant IL-10 did not affect the occurrence or degree of JHR [66,67]. Another possible inciting event for the hyper-inflammatory state of a JHR is the uptake of damaged spirochetes by large numbers of neutrophils and other phagocytes [63].

PROGNOSIS

The mortality rates for untreated tick-borne relapsing fever (TBRF) and louse-borne relapsing fever (LBRF) are in the ranges of 4 to 10 percent and 10 to 70 percent, respectively. With prompt antibiotic treatment, the mortality rate is reduced to less than 2 percent for TBRF and 2 to 5 percent overall for LBRF [68]. Among 72 cases of LBRF treated in Europe during 2010 to 2019, there was one death [21].

Features associated with a poor prognosis include:

- Stupor or coma on admission
- Diffuse bleeding
- Myocarditis
- Malnutrition
- Poor hepatic function
- Bronchopneumonia
- Coinfection with typhus, typhoid [69], or malaria [70,71]

Mortality secondary to relapsing fever has also been associated with Jarisch-Herxheimer reaction (JHR). In patients with LBRF, mortality associated with the JHR has been reported to be approximately 5 percent in the absence of adequate monitoring and resuscitation. (See 'Jarisch-Herxheimer reactions' above.)

After successful treatment, some patients may experience fatigue and other nonspecific symptoms for a few days to a few weeks. However, persistent infection causing chronic disease and/or disability after either natural recovery from or successful antibiotic treatment of relapsing fever has not been documented.

PREVENTION AND CONTROL

Reducing exposure — The best way to prevent relapsing fever is to decrease louse and tick exposure. There are no vaccines for either tick-borne relapsing fever (TBRF) or louse-borne relapsing fever (LBRF), and the prospects for a successful vaccine are poor because a single *Borrelia* strain can manifest any one of several different serotypic

identities when it first infects a mammalian host. Antibodies to one serotype might prevent infection with that particular serotype, but these antibodies would be ineffective against other serotypes. (See "Microbiology, pathogenesis, and epidemiology of relapsing fever".)

- Tick-borne relapsing fever – In rural areas of developing countries, the incidence of TBRF can be reduced by constructing houses with concrete or sealed plank floors and without thatched roofs or mud walls. Individuals are usually bitten by soft ticks while they are asleep in an infested dwelling.

  Log and other rustic cabins pose a particular risk in North America when rodents nest in the roofs or beneath the house or porch. Even structures built to modern standards can become a source of ticks once rodents have invaded attics and wall spaces.

  In residences and lodging posing a risk of infection, beds, preferably with metal legs, should be moved away from walls. In the past, buildings that were infested with *Ornithodoros* ticks were effectively treated with organophosphates (eg, malathion) and carbamate insecticides (eg, carbaryl). However, the use of these insecticides is becoming increasingly restricted, and they are being supplanted by less toxic pyrethroid insecticides, such as deltamethrin and cypermethrin.

- Louse-borne relapsing fever – LBRF can be prevented by lowering the risk of louse infestation through improved personal hygiene, reduced crowding, and improved access to washing facilities. Body lice live in clothes and feed on the skin of their host. A more in-depth discussion on delousing is found elsewhere. (See "Pediculosis corporis".)

Post-exposure prophylaxis for tick-borne relapsing fever — Post-exposure prophylaxis with either a single 100 mg dose of doxycycline or a short course of doxycycline (200 mg on day 1 followed by 100 mg daily for four days) is reasonable for individuals who are at high risk of infection following a suspected soft tick exposure in an endemic area. Antimicrobial therapy should be administered within four days of the suspected exposure. We prefer the longer duration in situations where there is a greater likelihood of a more severe or prolonged disease course (eg, splenectomized individuals or individuals with a B-cell immunodeficiency).

A suspected exposure to soft ticks consists of recently being in a high-risk setting (eg, an overnight stay in a mountain cabin, house, or lodge in North America where relapsing fever has recently occurred, or entering a cave where soft ticks are observed or relapsing fever was acquired by other visitors). There is usually no sign or experience of a bite. Post-exposure prophylaxis can also be used after an accidental inoculation with infected blood or culture medium in the laboratory, hospital, or clinic.

The efficacy of the five-day regimen was demonstrated in a randomized trial of 93 healthy individuals in Israel with a suspected tick exposure in a cave [72]. At three weeks, 10 cases of TBRF (defined as a patient with fever and a positive blood smear) were diagnosed; all were in the placebo group (attack rate 22 percent).

The single dose doxycycline regimen was subsequently supported in a retrospective study of spelunkers exposed to relapsing fever in caves in Israel [73]. Among 77 exposed individuals who received a 100 mg dose of doxycycline within three days of exposure, none developed TBRF, while there were 27 cases among 62 individuals without post-exposure prophylaxis.

There have been no trials of alternatives to tetracyclines for young children (eg, <8 years old) or for pregnant or nursing women. However, it is likely that four days of erythromycin or another macrolide would be equally efficacious. In addition, a single dose of doxycycline (but not other tetracyclines) is unlikely to have adverse effects for a young child or pregnant or nursing woman and would be an alternative to the short course of a macrolide.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Tick-borne infections (including Lyme disease, ehrlichiosis,

babesiosis, rickettsial infections, and others)".)

SUMMARY AND RECOMMENDATIONS

- **Etiology** – Relapsing fever is caused by spirochetes of the *Borrelia* genus. This is an arthropod-borne infection that occurs in two major forms: tick-borne (TBRF) and louse-borne (LBRF). (See 'Introduction' above.)

- **Clinical manifestations** – Relapsing fever presents with the sudden onset of fever, punctuated by an intervening afebrile period, and then recurrent fevers. The temperature is usually above 39°C and may be as high as 43°C. The first fever episode ends in a crisis phase that lasts approximately 15 to 30 minutes and consists of rigors, further elevations in temperature, and increases in pulse and blood pressure. (See 'Course of fever' above.)

  In addition to recurring fevers, patients can also present with nonspecific symptoms (eg, headache, myalgia, arthralgia, shaking chills, and abdominal complaints) and may be suspected of having sepsis. Patients can also develop neurologic, cardiac, pulmonary, or hematologic complications; the clinical presentation depends in part upon the form of infection (ie, TBRF versus LBRF). (See 'Other signs and symptoms' above.)

- **Evaluation and diagnosis** – Relapsing fever should be considered in patients who have both characteristic relapsing fevers and an epidemiologic history of exposure to body lice or soft-bodied ticks in geographic areas where LBRF or TBRF are endemic. In high-risk areas for TBRF or under circumstances such as a refugee camp with louse infestations, the initial episode of high fever should be suspected to be relapsing fever. (See 'Diagnostic approach' above.)

  To evaluate the patient for relapsing fever, thin and thick smears of blood should be performed to see if the organism can be identified. If the organism is not identified on smear, but there continues to be a high clinical suspicion for disease, polymerase chain reaction (PCR) testing can be performed. (See 'Diagnostic methods' above.)

- **Treatment** – Antimicrobial therapy is required for patients with a confirmed or suspected diagnosis of relapsing fever. The mortality rates for untreated TBRF and LBRF are in the ranges of 4 to 10 percent and 10 to 70 percent, respectively, but with prompt antibiotic treatment, the mortality rate is substantially reduced. (See 'Indications' above and 'Prognosis' above.)

  - **Preferred regimen for TBRF** – The majority of patients with TBRF present with severe symptoms and require hospitalization. For such patients, we suggest an intravenous (IV) beta-lactam, such as penicillin or ceftriaxone, rather than doxycycline for initial therapy (table 1) (**Grade 2C**). There have been decades of experience using penicillin to treat relapsing fever. (See 'Tick-borne relapsing fever' above.)

    - For patients without central nervous system (CNS) involvement, the duration of treatment for TBRF is generally 10 days. The patient can be switched to oral therapy when clinically stable.

    - For patients with CNS involvement and for pregnant persons, the duration of treatment should be extended for up to 14 days, and IV therapy with penicillin or ceftriaxone should be continued for the entire course.

  - **Alternative regimen for TBRF** – For patients with TBRF who cannot take a beta-lactam, a tetracycline (eg, doxycycline 100 mg twice daily orally or intravenously) should be used if possible. Macrolides (eg, azithromycin) are typically administered only if the patient cannot take a beta-lactam or a tetracycline, since there is less experience with these agents. (See 'Tick-borne relapsing fever' above and 'Special considerations for pregnant women and young children' above.)

  - **Regimen selection for LBRF** – Patients with LBRF should be treated with a single dose of intramuscular penicillin or a single dose of a tetracycline. For those unable to take one of these agents, a macrolide can be used. (See 'Louse-borne relapsing fever' above.)

- **Monitoring** – All patients who are treated for relapsing fever should be monitored for evidence of a Jarisch-Herxheimer reaction (JHR) after the first dose of therapy. Signs and symptoms of JHR include rigors, myalgias, further rise in the body temperature, increased respiratory rate, and hypotension lasting up to 24 hours. (See 'Jarisch-Herxheimer reactions' above.)

- **Prevention** – The best way to prevent relapsing fever is to decrease louse and tick exposures. In addition, post-exposure prophylaxis with an antibiotic (eg, doxycycline) can be administered following a suspected tick exposure in an endemic area. (See 'Prevention and control' above.)

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. Warrell DA. Louse-borne relapsing fever (Borrelia recurrentis infection). Epidemiol Infect 2019; 147:e106.
2. Lopez JE, Krishnavahjala A, Garcia MN, Bermudez S. Tick-Borne Relapsing Fever Spirochetes in the Americas. Vet Sci 2016; 3.
3. Dworkin MS, Schwan TG, Anderson DE Jr. Tick-borne relapsing fever in North America. Med Clin North Am 2002; 86:417.
4. Lucchini A, Lipani F, Costa C, et al. Louseborne Relapsing Fever among East African Refugees, Italy, 2015. Emerg Infect Dis 2016; 22:298.
5. Southern, P, Sanford, J. Relapsing fever: A clinical and microbiological review. Medicine (Baltimore) 1969; 48:129.
6. Mafi N, Yaglom HD, Levy C, et al. Tick-Borne Relapsing Fever in the White Mountains, Arizona, USA, 2013-2018. Emerg Infect Dis 2019; 25:649.
7. Lopez J, Hovius JW, Bergström S. Pathogenesis of Relapsing Fever. Curr Issues Mol Biol 2021; 42:519.
8. Osthoff M, Schibli A, Fadini D, et al. Louse-borne relapsing fever - report of four cases in Switzerland, June-December 2015. BMC Infect Dis 2016; 16:210.
9. Ayazi P, Mahyar A, Oveisi S, et al. Tick-borne Relapsing Fever in Children in the North-west of Iran, Qazvin. Prague Med Rep 2015; 116:193.
10. Cadavid D, Barbour AG. Neuroborreliosis during relapsing fever: review of the clinical manifestations, pathology, and treatment of infections in humans and experimental animals. Clin Infect Dis 1998; 26:151.
11. Zammarchi L, Antonelli A, Bartolini L, et al. Louse-Borne Relapsing Fever with Meningeal Involvement in an Immigrant from Somalia to Italy, October 2015. Vector Borne Zoonotic Dis 2016; 16:352.
12. Ellis L, Curtis MW, Gunter SM, Lopez JE. Relapsing Fever Infection Manifesting as Aseptic Meningitis, Texas, USA. Emerg Infect Dis 2021; 27.
13. Sciotto CG, Lauer BA, White WL, Istre GR. Detection of Borrelia in acridine orange-stained blood smears by fluorescence microscopy. Arch Pathol Lab Med 1983; 107:384.
14. Centers for Disease Control and Prevention (CDC). Acute respiratory distress syndrome in persons with tickborne relapsing fever--three states, 2004-2005. MMWR Morb Mortal Wkly Rep 2007; 56:1073.
15. Dennis DT, Awoke S, Doberstyn EB, Fresh JW. Bleeding in louse-borne relapsing fever in Ethiopia; clinical and laboratory features in 29 patients. East Afr Med J 1976; 53:220.
16. Perine PL, Parry EH, Vukotich D, et al. Bleeding in louse-borne relapsing fever. I. Clinical studies in 37 patients. Trans R Soc Trop Med Hyg 1971; 65:776.
17. Keller C, Zumblick M, Streubel K, et al. Hemorrhagic Diathesis in Borrelia recurrentis Infection Imported to Germany. Emerg Infect Dis 2016; 22:917.
18. Schuett J, Matkovic E. Soft tick relapsing fever in a traveler returning from Colorado. Blood 2024; 143:1429.
19. Ramos JM, Malmierca E, Reyes F, Tesfamariam A. Louse-borne relapsing fever in Ethiopian children: experience of a rural hospital. Trop Doct 2009; 39:34.

20. Centers for Disease Control and Prevention (CDC). Tickborne relapsing fever in a mother and newborn child--Colorado, 2011. MMWR Morb Mortal Wkly Rep 2012; 61:174.
21. Kahlig P, Paris DH, Neumayr A. Louse-borne relapsing fever-A systematic review and analysis of the literature: Part 1-Epidemiology and diagnostic aspects. PLoS Negl Trop Dis 2021; 15:e0008564.
22. Larsson C, Andersson M, Guo BP, et al. Complications of pregnancy and transplacental transmission of relapsing-fever borreliosis. J Infect Dis 2006; 194:1367.
23. Beeson AM, Kjemtrup A, Oltean H, et al. Soft Tick Relapsing Fever - United States, 2012-2021. MMWR Morb Mortal Wkly Rep 2023; 72:777.
24. Stoenner HG, Dodd T, Larsen C. Antigenic variation of Borrelia hermsii. J Exp Med 1982; 156:1297.
25. Magnarelli LA, Anderson JF, Johnson RC. Cross-reactivity in serological tests for Lyme disease and other spirochetal infections. J Infect Dis 1987; 156:183.
26. Larsson C, Bergström S. A novel and simple method for laboratory diagnosis of relapsing Fever borreliosis. Open Microbiol J 2008; 2:10.
27. Duray PH. The surgical pathology of human Lyme disease. An enlarging picture. Am J Surg Pathol 1987; 11 Suppl 1:47.
28. Park HK, Jones BE, Barbour AG. Erythema chronicum migrans of Lyme disease: diagnosis by monoclonal antibodies. J Am Acad Dermatol 1986; 15:406.
29. Dupont HT, La Scola B, Williams R, Raoult D. A focus of tick-borne relapsing fever in southern Zaire. Clin Infect Dis 1997; 25:139.
30. Brahim H, Perrier-Gros-Claude JD, Postic D, et al. Identifying relapsing fever Borrelia, Senegal. Emerg Infect Dis 2005; 11:474.
31. Halperin T, Orr N, Cohen R, et al. Detection of relapsing fever in human blood samples from Israel using PCR targeting the glycerophosphodiester phosphodiesterase (GlpQ) gene. Acta Trop 2006; 98:189.
32. Uhlmann EJ, Seed PC, Schwan TG, Storch GA. Tick-borne relapsing fever polymerase chain reaction of tick-borne relapsing fever caused by Borrelia hermsii. Pediatr Infect Dis J 2007; 26:267.
33. Elbir H, Henry M, Diatta G, et al. Multiplex real-time PCR diagnostic of relapsing fevers in Africa. PLoS Negl Trop Dis 2013; 7:e2042.
34. Kingry L, Sheldon S, Oatman S, et al. Targeted Metagenomics for Clinical Detection and Discovery of Bacterial Tick-Borne Pathogens. J Clin Microbiol 2020; 58.
35. Kelly R. Cultivation of Borrelia hermsi. Science 1971; 173:443.
36. Stoenner HG. Biology of Borrelia hermsii in Kelly medium. Appl Microbiol 1974; 28:540.
37. Barbour AG. Isolation and cultivation of Lyme disease spirochetes. Yale J Biol Med 1984; 57:521.
38. Barbour AG. Laboratory aspects of Lyme borreliosis. Clin Microbiol Rev 1988; 1:399.
39. Christensen J, Fischer RJ, McCoy BN, et al. Tickborne relapsing fever, Bitterroot Valley, Montana, USA. Emerg Infect Dis 2015; 21:217.
40. Schwan TG, Raffel SJ, Schrumpf ME, et al. Tick-borne relapsing fever and Borrelia hermsii, Los Angeles County, California, USA. Emerg Infect Dis 2009; 15:1026.
41. Schwan TG, Anderson JM, Lopez JE, et al. Endemic foci of the tick-borne relapsing fever spirochete Borrelia crocidurae in Mali, West Africa, and the potential for human infection. PLoS Negl Trop Dis 2012; 6:e1924.
42. Dworkin MS, Schwan TG, Anderson DE Jr, Borchardt SM. Tick-borne relapsing fever. Infect Dis Clin North Am 2008; 22:449.
43. Porcella SF, Raffel SJ, Schrumpf ME, et al. Serodiagnosis of Louse-Borne relapsing fever with glycerophosphodiester phosphodiesterase (GlpQ) from Borrelia recurrentis. J Clin Microbiol 2000; 38:3561.
44. Krause PJ, Narasimhan S, Wormser GP, et al. Human Borrelia miyamotoi infection in the United States. N Engl J

Med 2013; 368:291.

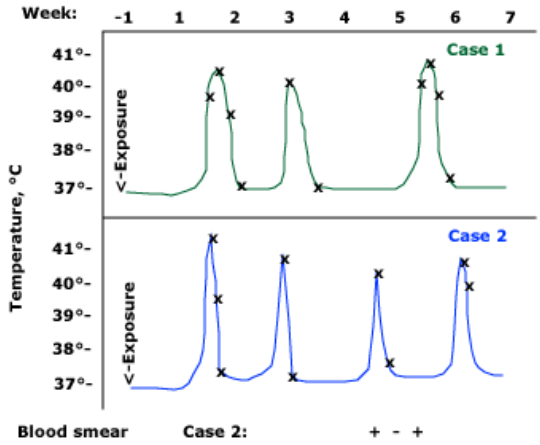
45. Molloy PJ, Telford SR 3rd, Chowdri HR, et al. Borrelia miyamotoi Disease in the Northeastern United States: A Case Series. Ann Intern Med 2015; 163:91.
46. Schwan TG, Schrumpf ME, Hinnebusch BJ, et al. GlpQ: an antigen for serological discrimination between relapsing fever and Lyme borreliosis. J Clin Microbiol 1996; 34:2483.
47. Barbour AG, Todd WJ, Stoenner HG. Action of penicillin on Borrelia hermsii. Antimicrob Agents Chemother 1982; 21:823.
48. Dever LL, Jorgensen JH, Barbour AG. Comparative in vitro activities of clarithromycin, azithromycin, and erythromycin against Borrelia burgdorferi. Antimicrob Agents Chemother 1993; 37:1704.
49. Kazragis RJ, Dever LL, Jorgensen JH, Barbour AG. In vivo activities of ceftriaxone and vancomycin against Borrelia spp. in the mouse brain and other sites. Antimicrob Agents Chemother 1996; 40:2632.
50. Horton JM, Blaser MJ. The spectrum of relapsing fever in the Rocky Mountains. Arch Intern Med 1985; 145:871.
51. CHERRY JK. The prevention and treatment of tick-borne relapsing fever with special reference to aureomycin and terramycin. Trans R Soc Trop Med Hyg 1955; 49:563.
52. Koetsveld J, Draga ROP, Wagemakers A, et al. In Vitro Susceptibility of the Relapsing-Fever Spirochete Borrelia miyamotoi to Antimicrobial Agents. Antimicrob Agents Chemother 2017; 61.
53. Butler T, Jones PK, Wallace CK. Borrelia recurrentis infection: single-dose antibiotic regimens and management of the Jarisch-Herxheimer reaction. J Infect Dis 1978; 137:573.
54. Perine PL, Krause DW, Awoke S, McDade JE. Single-dose doxycycline treatment of louse-borne relapsing fever and epidemic typhus. Lancet 1974; 2:742.
55. Perine PL, Teklu B. Antibiotic treatment of louse-borne relapsing fever in Ethiopia: a report of 377 cases. Am J Trop Med Hyg 1983; 32:1096.
56. Fekade D, Knox K, Hussein K, et al. Prevention of Jarisch-Herxheimer reactions by treatment with antibodies against tumor necrosis factor alpha. N Engl J Med 1996; 335:311.
57. Guerrier G, Doherty T. Comparison of antibiotic regimens for treating louse-borne relapsing fever: a meta-analysis. Trans R Soc Trop Med Hyg 2011; 105:483.
58. Vennila V, Madhu V, Rajesh R, et al. Tetracycline-induced discoloration of deciduous teeth: case series. J Int Oral Health 2014; 6:115.
59. Cohlan SQ, Bevelander G, Tiamsic T. Growth Inhibition of Prematures Receiving Tetracycline. Am J Dis Child 1963; 105:453.
60. Cross R, Ling C, Day NP, et al. Revisiting doxycycline in pregnancy and early childhood--time to rebuild its reputation? Expert Opin Drug Saf 2016; 15:367.
61. Cooper WO, Hernandez-Diaz S, Arbogast PG, et al. Antibiotics potentially used in response to bioterrorism and the risk of major congenital malformations. Paediatr Perinat Epidemiol 2009; 23:18.
62. Domínguez MC, Vergara S, Gómez MC, Roldán ME. Epidemiology of Tick-Borne Relapsing Fever in Endemic Area, Spain. Emerg Infect Dis 2020; 26:849.
63. Butler T. The Jarisch-Herxheimer Reaction After Antibiotic Treatment of Spirochetal Infections: A Review of Recent Cases and Our Understanding of Pathogenesis. Am J Trop Med Hyg 2017; 96:46.
64. Negussie Y, Remick DG, DeForge LE, et al. Detection of plasma tumor necrosis factor, interleukins 6, and 8 during the Jarisch-Herxheimer Reaction of relapsing fever. J Exp Med 1992; 175:1207.
65. Breuer A, Megged O, Kashat L, Assous MV. Quantitative real-time PCR in Borrelia persica tick-borne relapsing fever demonstrates correlation with the Jarisch-Herxheimer reaction. Eur J Clin Microbiol Infect Dis 2021; 40:1113.
66. Remick DG, Negussie Y, Fekade D, Griffin G. Pentoxifylline fails to prevent the Jarisch-Herxheimer reaction or associated cytokine release. J Infect Dis 1996; 174:627.

67. Cooper PJ, Fekade D, Remick DG, et al. Recombinant human interleukin-10 fails to alter proinflammatory cytokine production or physiologic changes associated with the Jarisch-Herxheimer reaction. J Infect Dis 2000; 181:203.
68. Barbour AG. Relapsing Fever. In: Harrison's principles of internal medicine, 21 Ed, Jameson JL, Fauci AS, Kasper D L, et al (Eds), McGraw Hill, New York City 2022.
69. ANDERSON TR, ZIMMERMAN LE. Relapsing fever in Korea; a clinicopathologic study of eleven fatal cases with special attention to association with Salmonella infections. Am J Pathol 1955; 31:1083.
70. Ramos JM, Reyes F, Tesfamariam A, Malmierca E. Louse-borne relapsing fever and malaria co-infection in Ethiopia. Trop Doct 2007; 37:121.
71. Nordstrand A, Bunikis I, Larsson C, et al. Tickborne relapsing fever diagnosis obscured by malaria, Togo. Emerg Infect Dis 2007; 13:117.
72. Hasin T, Davidovitch N, Cohen R, et al. Postexposure treatment with doxycycline for the prevention of tick-borne relapsing fever. N Engl J Med 2006; 355:148.
73. Binenbaum Y, Ben-Ami R, Baneth G, et al. Single Dose of Doxycycline for the Prevention of Tick-borne Relapsing Fever. Clin Infect Dis 2020; 71:1768.

Topic 7897 Version 20.0

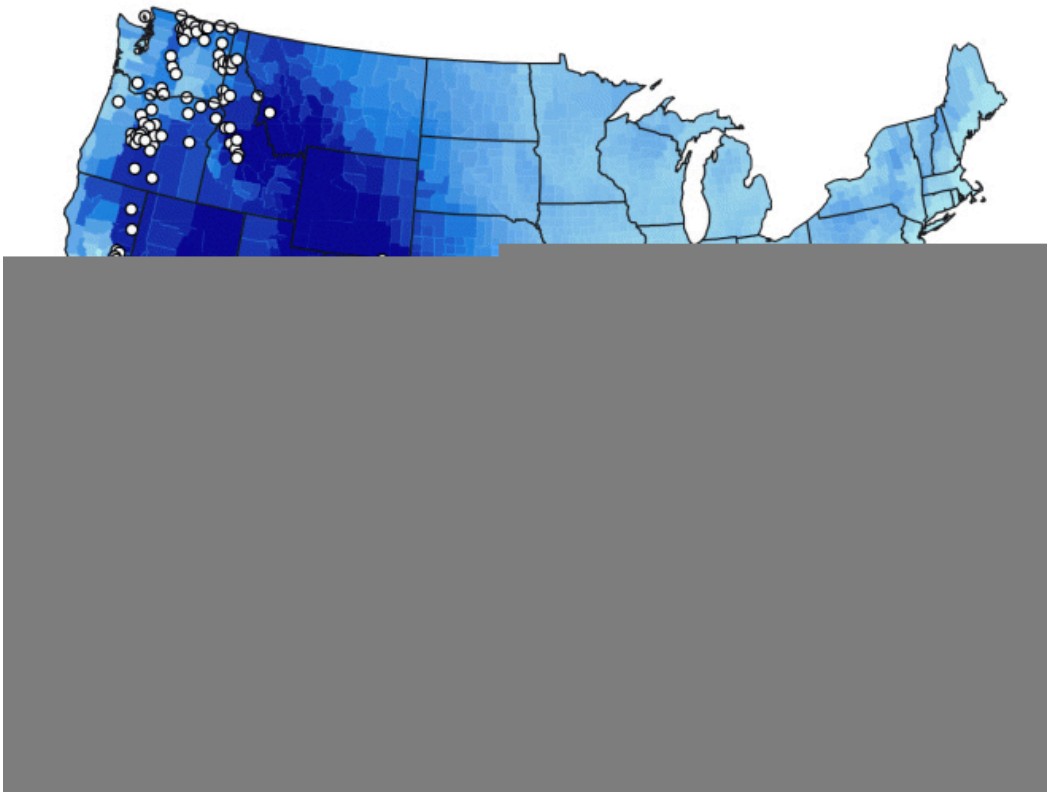
GRAPHICS

Figure 1: **Temperature pattern in two patients with relapsing fever**



Graphic 64476 Version 2.0

Figure 2: **Cases* of soft tick relapsing fever (n = 210), ¶ by county Δ of exposure — United States, 2012–2021**



* The figure does not show exact location of cases because they were arbitrarily placed within the county of exposure.

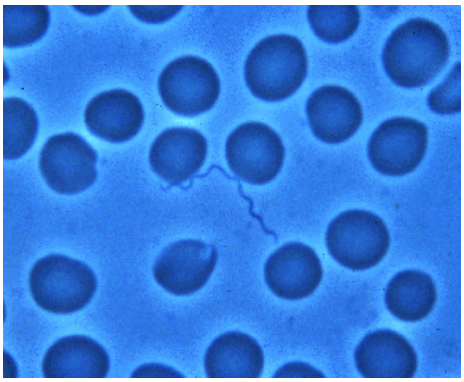
¶ Data on county of exposure was not available for all 251 cases included in this report.

Δ Mean elevation shown per county.

Reproduced from: Beeson AM, Kjemtrup A, Oltean H, et al. Soft tick relapsing fever - United States, 2012-2021. MMWR Morb Mortal Wkly Rep 2023; 72:777.

Graphic 116940 Version 3.0

Picture 1: **Thin smear in relapsing fever**



Wright stain reveals the spirochete of relapsing fever. The concentration of spirochetes in the blood must be 10^5 or greater to be visible in a single high power microscopic field.

Courtesy of Alan Barbour, MD.

Graphic 63059 Version 4.0

Table 1: **Antimicrobial agents for the treatment of tick-borne relapsing fever**

	Antimicrobial agent	Duration	Comment
Patients without known CNS disease			
Preferred regimen	**Adults:** ▪ Penicillin G (5 million units IV every six hours) **or** ceftriaxone (2 g IV daily or 1 g IV twice daily) **Children:** ▪ Penicillin G (50,000 to 100,000 units/kg IV every six hours, up to a maximum per dose of 5 million units) **or** ceftriaxone (50 to 75 mg/kg IV in a single dose, up to a maximum daily dose of 2 g)	10 days It is reasonable to switch to oral therapy when the patient is clinically stable*¶	There is more clinical experience with penicillin compared with ceftriaxone. However, ceftriaxone may be preferable for empiric therapy, if other infections (eg, meningococcal infection) are being considered, or for ease of administration.
Alternative regimen	Doxycycline PO/IV twice daily¶ A macrolide can be used for patients who cannot take a beta-lactam or doxycycline$^{\Delta}$	10 days	
Patients with CNS disease (eg, meningoencephalitis)$^{\diamond}$			
Preferred regimen	Penicillin G or ceftriaxone intravenously at the doses described above	10 to 14 days IV therapy with a penicillin or ceftriaxone should be continued for the entire duration	Refer to comments above about choice of penicillin or ceftriaxone.
Alternative regimen	All efforts should be made to administer penicillin or ceftriaxone If the patient is unable to tolerate a penicillin or ceftriaxone, a tetracycline (eg, doxycycline) is preferred, since tetracyclines have good CNS penetration¶	10 to 14 days	
Patients with isolated Bell's palsy			
Preferred regimen	Penicillin G or ceftriaxone intravenously at the doses described above	10 to 14 days It is reasonable to switch to an oral tetracycline (eg, doxycycline) when the patient is clinically stable¶	
Alternative regimen	Doxycycline PO/IV twice daily¶	10 to 14 days	
Pregnant women			
Preferred regimen	Penicillin G or ceftriaxone intravenously at the doses described above	10 to 14 days IV therapy with a beta-lactam antibiotic should be continued for the entire duration	IV penicillin or ceftriaxone should be continued regardless of the presence of CNS disease, given the greater risk of mortality for the mother and fetus or neonate.

Alternative regimen	All efforts should be made to administer a beta-lactam antibiotic If the patient is unable to tolerate a beta-lactam, the choice of an alternative agent depends upon the risks of therapy[§]	10 to 14 days	

All patients being treated for TBRF should be observed for a Jarisch-Herxheimer reaction, which typically occurs within several hours of the first dose.

CNS: central nervous system; JHR: Jarisch-Herxheimer reaction; PO: orally; IV: intravenous; TBRF: tick-borne relapsing fever.
* If possible, we prefer to switch the patient to an oral tetracycline (eg, doxycycline), as oral tetracyclines have good CNS penetration. However, if there are contraindications to tetracyclines, an oral beta-lactam, such as penicillin or a third generation cephalosporin, can be used. We avoid amoxicillin due to conflicting data about the efficacy of this agent.
¶ The dose of doxycycline for adults is 100 mg twice daily. For children <45 kg, the dose of doxycycline is 2.2 mg/kg/dose (maximum dose: 100 mg/dose) every 12 hours. If doxycycline is not available, tetracycline (eg, 500 mg four times daily PO for adults) can be used for many nonpregnant patients. However, for nursing mothers and children <8 years old, a macrolide, such as erythromycin or azithromycin, may be preferred if doxycycline cannot be used. Although the risk of dental staining is very low for children when a short course of doxycycline is administered, the risk of toxicity appears greater if other tetracyclines are used, and the choice of agent must be determined on a case-by-case basis. Refer to the topic that discusses the management of relapsing fever for additional information on the treatment of children.
Δ There is most experience with erythromycin base (500 mg PO every six hours for adults; 12.5 mg/kg [maximum dose 500 mg] PO every six hours for children) as an alternative to tetracyclines; however, this agent is likely to produce adverse drug effects. A macrolide such as azithromycin is better tolerated and is likely to be as effective at doses used to treat other *Borrelia* infections (eg, 500 mg daily for adults; 10 mg/kg [maximum dose 500 mg per day] daily for children).
◇ Refer to the topic that discusses the clinical manifestations of relapsing fever for more information on neurologic signs and symptoms.
§ For pregnant women who are unable to tolerate a beta-lactam, the choice of an alternative agent depends upon the risks of therapy. Available data suggest that doxycycline has not been associated with serious adverse events. However, the risk appears greater if other tetracyclines are used, and if doxycycline is not available, a macrolide may be preferred, particularly in those without evidence of CNS disease. Refer to the topic that discusses management of relapsing fever for additional information on the treatment of pregnant women.

Graphic 116192 Version 2.0

Contributor Disclosures

Alan G Barbour, MD Grant/Research/Clinical Trial Support: National Institutes of Health [Infectious disease pathogenesis]. Consultant/Advisory Boards: National Institutes of Health [Microbiology]. All of the relevant financial relationships listed have been mitigated. **Daniel J Sexton, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Sheldon L Kaplan, MD** Grant/Research/Clinical Trial Support: Pfizer [Streptococcus pneumoniae]. Other Financial Interest: Elsevier [Textbook honoraria – Pediatric infectious diseases]. All of the relevant financial relationships listed have been mitigated. **Jennifer Mitty, MD, MPH** No relevant financial relationship(s) with ineligible companies to disclose.

Contributor disclosures are reviewed for conflicts of interest by the editorial group. When found, these are addressed by vetting through a multi-level review process, and through requirements for references to be provided to support the content. Appropriately referenced content is required of all authors and must conform to UpToDate standards of evidence.

Conflict of interest policy

→